

Jeremy Lacomis

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About Me

I hold a Ph.D. from Carnegie Mellon University and have a deep interest in automated software engineering. My expertise lies in decompilers and low-level source code analysis, where I have leveraged machine learning techniques to automatically enhance decompiler output, making it more accessible and understandable for reverse engineers. I am passionate about applying my skills and experience to solve real-world problems and am seeking opportunities to contribute to impactful projects in this domain.

Quick Guide

- Deep understanding of compilation, low-level programming, and decompilation.
- Experience combining machine learning with insights to lower the cognitive load of reverse-engineers.
- Comfortable with Python, C, Java, OCaml, and x86-64 assembly language.

Education

PhD	Carnegie Mellon University , Software Engineering	2017 - 2023
	• Dissertation: Automatically Annotating Decompiled Code with Meaningful Names and Types • Advisors: Claire Le Goues and Bogdan Vasilescu	
BA	University of Virginia , Computer Science	2015 - 2017
AS	Piedmont Virginia Community College , Computer Science	2010 - 2015

Experience

Carnegie Mellon University , Mark Stehlik Postdoctoral Teaching Fellow	Pittsburgh, PA June 2023 – July 2025
• Instructor of record for four semesters of an introductory software engineering course. • Designed new homework assignments, labs, and exams from scratch. • Organized and led teams of 10 TAs with up to two other co-instructors.	
GammaTech , Software Engineer Intern	Ithaca, NY May 2019 – Aug 2019
• Completely refactored the Python interface to the GammaTech Intermediate Representation for Binaries (GTIRB), adding documentation and automated tests. • Added functionality to the Python interface for serializing and deserializing GTIRB tuples, vectors, and <code>intt64_t</code> types. • Did preliminary experimentation with GTIRB to identify ways to automatically reduce the attack surface of binaries without access to their source code.	

Selected Publications

A Human Study of Automatically Generated Decompiler Annotations	DSN 2025 (accepted)
Yuwei Yang, Sykler Grandel, Jeremy Lacomis , Edward J. Schwartz, Bogdan Vasilescu, Claire Le Goues, Kevin Leach	
• Performed a human study of 40 reverse-engineers to evaluate the effectiveness of automated decompiler annotations. • Empirically demonstrated a lack of correlation between current best-known ML performance metrics and the effectiveness of variable type and name recovery.	

- Found that users prefer code with annotations over code without annotations, even when the annotations do not significantly increase accuracy or reduce the time taken to understand decompiled code.

A Taxonomy of C Decompiler Fidelity Issues

USENIX 2024

Luke Dramko, **Jeremy Lacomis**, Edward J. Schwartz, Claire Le Goues, Bogdan Vasilescu

- Created a comprehensive, hierarchical taxonomy of fidelity issues in decompiled code beyond names and types
- Compared the effectiveness of four different modern decompilers

VarCLR: Variable Semantic Representation Pre-training via Contrastive Learning

ICSE 2022

Qibin Chen, **Jeremy Lacomis**, Edward J. Schwartz, Claire Le Goues, Graham Neubig, Bogdan Vasilescu

- Presented VarCLR, a novel method based on contrastive learning that learns general-purpose variable semantic representations
- Created GITHUBRENAMES, a weakly-supervised dataset of variable renamings collected directly from GITHUB

Augmenting Decompiler Output with Learned Variable Names and Types

USENIX 2022

Distinguished Paper Award

Qibin Chen, **Jeremy Lacomis**, Edward J. Schwartz, Claire Le Goues, Graham Neubig, Bogdan Vasilescu

- Introduced the **DecompI**ler variable **ReTY**per (DIRTY), a Transformer-based neural network model designed to recover both semantic and syntactic variable types in decompiled code
- Collected the **Dataset for Idiomatic ReTyping** (DIRT), a dataset of nearly 1 million public C functions for training models for renaming and retyping variables

DIRE: A Neural Approach to Decompiled Identifier Naming

ASE 2019

Jeremy Lacomis, Pengcheng Yin, Edward J. Schwartz, Miltiadis Allamanis, Claire Le Goues, Graham Neubig, Bogdan Vasilescu

- Introduced the **Decompiled Identifier Renaming Engine** (DIRE), a technique for assigning meaningful names to decompiled variables using neural network techniques
- Presented a novel technique for generating corpora suitable for training both lexical and graph-based probabilistic models of variable names in decompiled code

Projects

DIRTY

github.com/CMUSTRUDEL/DIRTY 

- A Transformer-based neural network model designed to recover both semantic and syntactic variable types in decompiled code
- Configurable to also predict variable names at the same time
- Tools Used: Python, PyTorch, Hex-Rays

DIRE

github.com/CMUSTRUDEL/DIRE 

- A tool for assigning meaningful names to decompiled variables using neural network techniques
- Tools Used: Python, PyTorch, Hex-Rays

Professional Service and Affiliations

Selection Committee Chair, REUSE@CMU	2019 - 2023
Mentor, SCS Graduate Application Support Program	2020
Member, ISR DEI Committee	2020 - 2023
Journal Referee, Automated Software Engineering	2024

References

Claire Le Goues, Professor of Computer Science: Carnegie Mellon University
clegoues@andrew.cmu.edu
Zachary Fry, Senior Scientist III: GrammaTech
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